Ajay Wadhwani

617-767-1605 | ajaywa@umich.edu | linkedin.com/in/ajay-wadhwani2006/ | github.com/ajayw36/

Education

University of Michigan — Ann Arbor, MI

B.S. Honors Mathematics and Computer Science — *Graduating May 2029 — GPA: 3.97*

- **Relevant Coursework**: Data Structures and Algorithms, Honors Analysis, Linear Algebra, Discrete Math, Combinatorics, Probability Theory, Computer Organization

Experience

Incoming Software Engineering Intern — Summer 2027

MongoDB — *New York, NY*

- Incoming intern contributing to MongoDB's database technologies and infrastructure.

AI Engineering Intern — May 2026 – Aug. 2026

Blitzy — *Boston, MA*

- Built a demo catalog for AI solutions consultants, reducing demo preparation time and sales cycle length.
- Evaluated Blitzy on frontier benchmarks, including SWE-bench Pro, SWE-Atlas, and DeepSWE, and built internal evaluation frameworks for accuracy, token usage, runtime, and cost, which were used to improve the platform.

Research Assistant — May 2026 – Aug. 2026

Thurber Lab, University of Michigan Chemical Engineering — *Ann Arbor, MI*

- Supported ADC cancer-therapeutics research using C++ agent-based simulations and Python workflows.
- Built Python/Bash pipelines to convert Great Lakes ensemble outputs into structured CSVs and visualizations.

Director of Computer Science — Sep. 2023 – May 2025

Greater Boston STEM Program (gbSTEM) — *Boston, MA*

- June 2024 - May 2025 (CS Director): Managed a team of 15 instructors and worked with curriculum developers to create an engaging project-based curriculum for all of the Scratch, Python, and Web Development classes.
- September 2023 - May 2024 (Python Instructor): Taught online programming classes to a group of middle school students twice a week, guiding them through core programming concepts and hands-on projects.

Machine Learning Intern — July 2024 – Sep. 2024

Mass General Hospital — *Boston, MA*

- Developed and validated ML models in Python to identify sharp-wave ripples, an important biomarker in epilepsy research, from high-density EEG datasets (∼10k multi-channel recordings, sampled at 1250Hz).
- Authored a 2,000-word paper detailing data processing and analysis, and presented results to faculty and peers.

Projects

KTP App Development Committee Lead | *Flutter SDK, Dart, Firebase, Figma* — Sep. 2025 – Present

- Designed and implemented a member-tracking feature (20+ pages — iOS, Android and Web) using Flutter SDK and Dart as part of a collaborative development team for *Kappa Theta Pi*, a professional technology fraternity.
- Optimized backend storage for 700+ users in Google Firebase, and created UI prototypes in Figma.

Healthcare Analytics Dashboard | *React, Spring Boot, Java, SQL, Tableau, TypeScript* — Sep. 2025 – Dec. 2025

- Built a full-stack web dashboard with a React frontend and Spring Boot backend for *Integrate Healthcare*.
- Created data pipelines and embedded Tableau visualizations to display hospital billing and operations data.

Limit Order Book Simulator | *C++* — Jan. 2026 – Feb. 2026

- Built a discrete-event limit order book simulator implementing price-time priority matching with full support for limit, market, and cancel orders, including partial fills and dynamic book state updates.
- Processed 1M+ timestamped events to evaluate trading strategies through slippage and fill probability analysis.

Technical Skills

Languages: Python (NumPy, Pandas, Matplotlib, Scikit-learn), C++, Java, JavaScript (React, Next.js), Dart
Frameworks/Tools: Git, LaTeX, Flutter SDK, Spring Boot, Firebase

Other

Activities/Programs: Kappa Theta Pi, Professional Technology Fraternity (Coding Committee Lead), Michigan Investment Group (Quantitative Trader), MongoDB Next in Tech Summit (Selected Participant) — Summer 2026

Awards and Honors: William J. Branstrom Freshman Prize (Ranked in top 5% of class), Michigan Quant Conference (Algorithmic trading strategy and trading games)